

PIE

Platform Intelligence Engine

Functional Documentation

Understanding Azure Data Factory before making changes.

Version:	1.0
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Prepared Date:	August 30, 2026
Prepared By:	Project Team
Organization:	Azure Data Estate Operations

Platform Intelligence Engine (PIE)

AI-Powered Engineering Intelligence Layer

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4. Project Overview

Platform Intelligence Engine (PIE) is an AI-powered Engineering Intelligence and Knowledge Layer designed for Microsoft Azure Data Factory (ADF), with a multi-year roadmap extending to the complete Azure Data Estate.

The Core Problem It Solves

Azure Data Factory environments in modern enterprises grow rapidly, often containing hundreds of pipelines, datasets, linked services, triggers, and mapping data flows. As complexity scales, organizations face several critical challenges:

1. **Tribal Knowledge Loss:** The understanding of how and why pipelines connect resides within a few individual developers. If they leave, that knowledge is lost.
2. **Outdated Documentation:** Manual documentation degrades immediately after implementation because environments change rapidly.
3. **Manual Dependency Analysis:** Engineers spend hours opening JSON files, tracking dataset schemas, and assessing what assets are impacted by a schema change.
4. **Slow Onboarding:** New team members take weeks to map out the pipeline topology and understand business workflows.
5. **High Production Change Risk:** Modifying or deleting a dataset or linked service may cause unintended cascading breakages across downstream data pipelines.

Purpose and Business Value

PIE bridges the gap between raw metadata optimized for execution and human understanding. It does **NOT** execute or edit pipelines, but runs completely out-of-band to provide decision support.

- **De-risks Changes:** Instantly maps out the upstream and downstream blast radius of any modification.
- **Identifies Technical Debt:** Scans the metadata to locate unused "orphan" pipelines, fragile activities with zero retry policies, and duplicated parameters.
- **Accelerates Productivity:** Provides instant natural language summaries, architecture audits, and modernization code migrations (e.g. PySpark or dbt code).
- **Reduces Onboarding Time:** Enables new developers to explore the entire environment using a clean dashboard and an AI-assisted copilot.

Target Personas

- **Data Engineers:** Daily pipeline understanding, dependency navigation, and risk mitigation.
- **Technical Leads:** Architecture reviews, risk assessment of incoming changes, and standard enforcement.
- **Solution Architects:** End-to-end platform design validation, integration modeling, and technical debt audit.
- **Support / Operations Engineers:** Upstream/downstream root-cause investigation during incident resolution.

Simple High-Level Workflow

The entire system operates in a simple, secure, read-only lifecycle:

1. **Authentication:** The user logs in using Microsoft Entra ID OAuth2 PKCE.
2. **Discovery & Sync:** PIE calls Azure ARM APIs to discover data factories and sync metadata under Reader permissions.
3. **Knowledge Graph Compilation:** Synced metadata is normalized and modeled as a directed knowledge graph.
4. **Interactive Explorer & Sandbox:** The user browses the dashboard metrics, drills into data canvases, and runs "What-If" deletion risk simulations.
5. **Context Building & Token Budgeting:** When a user queries the system, PIE extracts a local subgraph, compresses it by 95%+, and compiles it into a prompt template.

6. **AI Reasoning:** The grounded prompt is processed by an LLM (Gemini, LLaMA, or Azure OpenAI) to generate zero-hallucination insights.

5. Scope

The scope of the Platform Intelligence Engine (PIE) is structured to provide high value while maintaining strict boundaries to secure the environment.

In Scope (MVP)

- **Azure Data Factory Assets:** Fully supports pipelines, activities, datasets, linked services, triggers, global parameters, and mapping data flows.
- **Metadata Discovery:** Securely crawls selected subscriptions and resource groups to read JSON configurations.
- **Knowledge Graph Compilation:** Builds an in-memory graph containing vertices (assets) and directed edges representing relationships (**USES**, **READS**, **WRITES**, **CALLS**, **EXECUTES**, **TRIGGERED_BY**, **DEPENDS_ON**).
- **What-If Deletion Simulation:** Deterministically lists broken readers, broken writers, affected pipelines, and generates step-by-step remediation plans.
- **Change Impact Analysis:** Rates the risk (CRITICAL, HIGH, MEDIUM, LOW) of deletions, modifications, and renames across the workspace.
- **Enterprise Governance Audits:** Mapped SaaS vendor endpoints, concurrency schedule collisions, zero-retry fragile activities, and duplicate global parameters.
- **AI Reasoning & Conversational Chat:** Context-grounded Q&A, business summaries, technical summaries, and PySpark/dbt code migrations.
- **Markdown Export:** Generates Git-ready Markdown files for documentation.

Out of Scope (Future Roadmap)

- Databricks, Fabric, Synapse Analytics, Purview, and DevOps/Git integrations (deferred to horizontal releases).
- Editing, deploying, deleting, or executing Azure Data Factory resources (PIE is strictly out-of-band).
- Runtime monitoring, execution logs, and historical pipeline runs.
- Cloud cost optimization.

Prerequisites and Dependencies

- **Azure Reader Role:** The user or service principal must have at least the Reader role to query ARM APIs.
 - **API Credentials:** Google Gemini, Azure OpenAI, or NVIDIA NIM credentials configured in Settings (or default to offline Mock mode).
 - **Entra ID Redirect URL:** Callback server must be reachable on port 8100.
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6. User Roles and Access

Access control in PIE aligns with the read-only principles of the system, leveraging existing Azure RBAC permissions while defining user profiles.

Role Name	Who Uses	Permissions & Views	Restrictions
---	---	---	---
Data Engineer	Daily developers, ETL designers	Full catalog browsing, dependency visualization, chat copilot Q&A, PySpark code migrations, what-if deletion simulations.	Cannot modify system settings or update plain-text API keys.
Technical Lead & Architect	Engineering leads, Solution designers	All Data Engineer views, plus settings modification, API key entry, technical debt reports, and concurrency heatmaps.	No restrictions; has full administrative configuration access.
Support & Operations	Maintenance engineers, Incident managers	Catalog browsing, lineage graphs, and search capabilities to trace dependencies and troubleshoot failures.	Cannot view technical debt reports, change settings, or access AI code migrations.

7. Functional Modules

PIE consists of seven core modules that work together to deliver engineering intelligence.

Module 1: Authentication & Session Management

- **Purpose:** Securely logs users in using Microsoft Entra ID OAuth2 PKCE without storing credentials.
- **Access:** All user roles.
- **Entry Point:** Initial page load (forces redirection to Login if no token exists).
- **Screens Involved:** Login page, Microsoft Entra ID popup.
- **Main Actions:** Clicking "Login with Microsoft".
- **Inputs:** Entra ID credentials entered into the Microsoft secure popup.
- **Validations:** Generates random `state`, `code_verifier`, and `code_challenge`. Rejects states older than 10 minutes to prevent replay attacks.
- **System Response:** Catch redirect on Port 8100, exchange code for access token, decode JWT claims (tid, upn, name), store token in-memory, and return a unique session token to the browser.
- **Outputs:** `x_session_token` stored in browser `localStorage`.
- **Exit Point:** Setup Wizard Step 1 (Subscriptions).

Module 2: Discovery & Sync Engine (Ingestion)

- **Purpose:** Pulls metadata from Azure ARM and populates the database cache and Knowledge Graph.
- **Access:** Data Engineers, Technical Leads.
- **Entry Point:** Setup Wizard Step 1 & 2.
- **Screens Involved:** Setup Wizard, Dashboard.
- **Main Actions:** Select subscription, select Data Factory, click Sync.
- **Inputs:** Target subscription and factory selections.
- **Validations:** Verifies permissions; validates factory resource group.
- **System Response:** Extracts pipelines, activities, datasets, linked services, and triggers. Sanitizes secret keys, stores data in the local cache, compiles the Knowledge Graph, and saves metadata to disk cache.
- **Outputs:** Populates dashboard statistics and sidebar catalog.
- **Exit Point:** Redirect to Workspace Dashboard.

Module 3: Knowledge Graph Explorer & Data Canvas

- **Purpose:** Compiles the normalized metadata into a directed dependency graph and visualizes it.
- **Access:** All roles.
- **Entry Point:** Workspace dashboard stat cards or sidebar menu.
- **Screens Involved:** Dashboard, Data Canvas.
- **Main Actions:** Click a statistic card (e.g. Pipelines) to drill down, select a pipeline to view its canvas, and inspect activity execution dependencies.
- **Inputs:** Clicking items in the list.
- **Validations:** Verifies the selected asset exists in the metadata repository.
- **System Response:** Computes upstream/downstream lineage, loads parameters/variables, and renders a table of pipeline activities showing `dependsOn` relations.

- **Outputs:** Detailed pipeline specification and visual data canvas.
- **Exit Point:** Navigating back to overview or triggering impact analysis.

Module 4: Change Impact & Deletion Simulator

- **Purpose:** Predicts downstream breakages and calculates risk scores for proposed changes.
- **Access:** Data Engineers, Technical Leads.
- **Entry Point:** Pipeline detail page or dataset catalog.
- **Screens Involved:** Change Impact Panel.
- **Main Actions:** Select change action (e.g. DELETE, MODIFY) and click "Analyze Impact".
- **Inputs:** Target asset name and change action.
- **Validations:** Checks if the asset exists. Triggers disambiguation if a pipeline and dataset share the same name.
- **System Response:** Traverses the graph to trace downstream consumers, calculates risk score (0-100) and risk level, lists direct/indirect failures, and suggests remediation steps.
- **Outputs:** Change Impact report and recommendations.
- **Exit Point:** Review and export findings.

Module 5: Enterprise Governance & Technical Debt Scan

- **Purpose:** Evaluates metadata for compliance, naming standards, zero-retry fragility, schedule collisions, and SaaS connections.
- **Access:** Technical Leads, Architects.
- **Entry Point:** Dashboard overview.
- **Screens Involved:** Dashboard Insights.
- **Main Actions:** View insights cards, examine orphan pipelines, and evaluate peak schedule concurrency.
- **Inputs:** Executed automatically on synchronization.
- **Validations:** Checks pipeline incoming edges (orphans), activity retry counters (zero-retries), and overlapping cron expressions (concurrency).
- **System Response:** Flags compliance issues and updates stats.
- **Outputs:** Lists of technical debt items and SaaS vendors.
- **Exit Point:** Refactoring pipeline topologies.

Module 6: Conversational Engineering Assistant (AI Chat)

- **Purpose:** AI-powered drawer for natural language discovery, pipeline explanation, and PySpark code migrations.
- **Access:** All roles (code generation restricted to Engineers/Leads).
- **Entry Point:** Left Pane 2 (Chat sidebar).
- **Screens Involved:** Main Workspace (Sidebar Chat).
- **Main Actions:** Type questions and select AI model.
- **Inputs:** Natural language queries.
- **Validations:** Intent routing classifies query. SchemaCompressor compresses JSON by up to 99% to fit LLM limits.
- **System Response:** Grounded prompt sent to LLM provider; streams response to UI.
- **Outputs:** Real-time streamed Markdown explanations and code blocks.
- **Exit Point:** Copying code or asking follow-up questions.

Module 7: Settings & API Key Configuration

- **Purpose:** Updates API credentials for AI providers and persists them across server restarts.

- **Access:** Technical Leads, Architects.
 - **Entry Point:** Sidebar gear icon.
 - **Screens Involved:** Settings Modal.
 - **Main Actions:** Enter API keys and click "Save Keys".
 - **Inputs:** Text strings for Gemini, Azure OpenAI, and NVIDIA keys.
 - **Validations:** Saves values to `os.environ` and `.env` file without duplication.
 - **System Response:** Re-initializes LLM client providers.
 - **Outputs:** Updated `.env` configuration file.
 - **Exit Point:** Close modal.
-

8. Detailed User Workflows

This section outlines the step-by-step user workflows for primary operations in PIE.

Workflow Name: User Login and Session Initialization

- **Purpose:** Authenticate the user against Entra ID and establish a secure, read-only session.
- **User Role:** All Roles.
- **Preconditions:** Server is running; user has an active Entra ID account.
- **Trigger:** Opening the application URL in a web browser.
- **Step-by-Step Flow:**
 1. System checks for `x_session_token` in `localStorage`. Finding none, it redirects to the Setup Wizard Step 0.
 2. The user clicks the "Login with Microsoft" button.
 3. System opens a secure browser popup window pointing to the Entra ID authorization login URL.
 4. User enters credentials and authenticates successfully in the popup.
 5. Microsoft redirects the popup to the local callback listener on Port 8100, passing the auth code.
 6. Backend exchanges the code, decodes JWT claims, registers the session, and closes the popup.
 7. Frontend receives a `postMessage` (or poll complete) containing the session token.
- **Expected Result:** The user is authenticated; session token is stored.
- **Postconditions:** User is redirected to Step 1 (Select Subscription).
- **Exceptions / Alternate Flows:** If login fails or is cancelled, the popup closes, and the Setup Wizard displays a red authentication error message.
- **Business Rules:** PKCE parameters must be validated; logins expire after the token TTL.
- **Related Screens or Modules:** Setup Wizard, Module 1 (Authentication).

Workflow Name: Azure Data Factory Selection and Sync

- **Purpose:** Select the target Azure subscription and Data Factory instance, and ingest its metadata.
- **User Role:** Data Engineer, Technical Lead.
- **Preconditions:** Successful authentication.
- **Trigger:** Completion of the login workflow.
- **Step-by-Step Flow:**
 1. Setup Wizard fetches and lists accessible subscriptions (Step 1).
 2. The user clicks on the desired subscription.
 3. System lists all factories discovered within that subscription (Step 2), showing their sync status.
 4. The user clicks the desired Data Factory.
 5. System triggers a sync request (`POST /discovery/sync`), displaying a "Syncing..." loading spinner (Step 3).
 6. The backend pulls factory JSON definitions, normalizes them, builds the graph, and updates cache.
 7. Upon completion, a checkmark and "Ready to Explore" screen is shown (Step 4).
 8. User clicks "Enter Workspace" to load the dashboard.
- **Expected Result:** Metadata is ingested and indexed, and the main workspace opens.
- **Postconditions:** Active factory name is stored in `localStorage` as `selected_factory`.
- **Exceptions / Alternate Flows:** If ARM extraction fails (e.g. network failure), the backend falls back to creating a mock data

factory template so testing can continue, and logs a warning.

- **Business Rules:** Discovery requires at least Reader access; sync overrides existing in-memory graph.
- **Related Screens or Modules:** Setup Wizard, Dashboard, Module 2 (Discovery).

Workflow Name: Deletion Risk Simulation & Change Impact Analysis

- **Purpose:** Analyze the cascading blast radius of deleting or modifying an ADF resource.
- **User Role:** Data Engineer, Technical Lead.
- **Preconditions:** Active synced factory.
- **Trigger:** Clicking "Impact Analysis" from a pipeline canvas or dataset catalog.
- **Step-by-Step Flow:**
 1. The user navigates to the Change Impact Panel for the target asset (e.g., **DataLakeCsv**).
 2. The user selects a change type (e.g., DELETE) from the dropdown list.
 3. The user clicks the "Analyze Impact" button.
 4. System traverses the directed graph to identify all upstream producers and downstream consumers.
 5. The risk engine calculates the numerical risk score and risk level (CRITICAL, HIGH, MEDIUM, LOW).
 6. Direct and indirect impacts are mapped, outlining exactly which pipelines and activities will break.
 7. The system generates potential consequences and AI-driven migration recommendations.
- **Expected Result:** A detailed Change Impact report is generated and displayed on the screen.
- **Postconditions:** Findings are shown; user can review details of affected pipelines.
- **Exceptions / Alternate Flows:** If the asset name matches multiple resources (e.g. a pipeline and dataset have the same name), a yellow "Disambiguation Notice" warns the user.
- **Business Rules:** Deleting datasets used by copy activities constitutes a High/Critical risk.
- **Related Screens or Modules:** Change Impact Panel, Module 4 (Change Impact).

Workflow Name: AI Chat Deep Dive and Code Modernization

- **Purpose:** Consult the AI assistant for architectural explanation or PySpark/dbt code migrations.
- **User Role:** Data Engineer, Technical Lead.
- **Preconditions:** Synced factory; LLM API keys configured.
- **Trigger:** Entering a question in the Chat drawer.
- **Step-by-Step Flow:**
 1. The user selects an AI provider (e.g., Google Gemini) from the Sidebar dropdown.
 2. The user opens the Chat drawer and types: "Generate PySpark code for Load_LeadToCash_Charges".
 3. The backend Query Router detects the **CODE_GEN** intent and extracts the pipeline name.
 4. The Context Builder extracts the metadata schema for **Load_LeadToCash_Charges** and compresses it by 95%+.
 5. The prompt is injected into the code generation template and sent to the LLM.
 6. The AI assistant returns PySpark DataFrame code, structural mapping, and an explanation.
- **Expected Result:** The user receives a detailed code migration response in Markdown format.
- **Postconditions:** The user can copy the code snippet or ask follow-up questions.
- **Exceptions / Alternate Flows:** If the selected LLM provider API key is missing, the system falls back to the Deterministic Mock Provider and writes a warning log.
- **Business Rules:** LLM responses must be grounded strictly within the extracted metadata context.
- **Related Screens or Modules:** Chat Drawer, Module 6 (AI Chat).

9. Screen/Page-wise Documentation

PIE features a clean, responsive single-page visual layout divided into a sidebar settings column, a chat panel, and a dynamic visualization canvas.

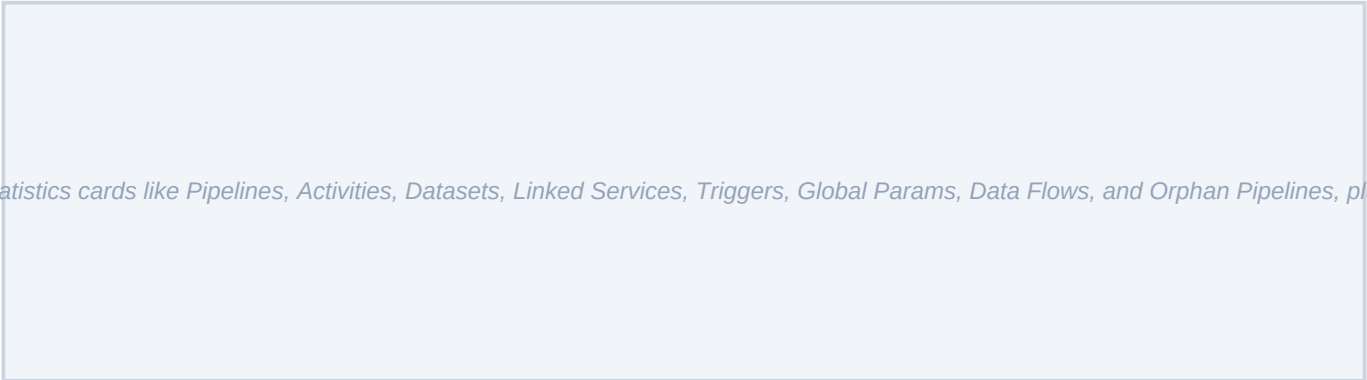
Screen 1: Setup Wizard (Login & Sync Wizard)

- **Purpose:** Authenticate user session and ingest factory assets.
- **User Access:** All users.
- **Navigation:** Root URL (/) when no valid session is found.
- **Visual Description:** A centered card layout showing the PIE logo and a progress bar at the top, transitioning through five states:
 1. **Login State (Step 0):** Contains a description and a "Login with Microsoft" button.
 2. **Subscription Selection (Step 1):** Lists discovered Azure subscriptions in vertical cards.
 3. **Factory Selection (Step 2):** Lists data factories in the selected subscription, detailing their resource group and sync state.
 4. **Syncing Progress (Step 3):** Shows a rotating circular spinner and a status message.
 5. **Ready Page (Step 4):** Displays a green success checkmark and an "Enter Workspace" button.
- **Screenshot Placeholder:** [Screenshot to be added: Setup Wizard Screen]
- **Field Details:**

Field/Button	Type	Mandatory	Description
---	---	---	---
Login with Microsoft	Button	Yes	Opens OAuth2 authentication popup window.
Subscription list	Grid Cards	Yes	Clickable list of accessible Azure subscriptions.
Factory list	Grid Cards	Yes	Clickable list of Azure Data Factories with RG and sync indicator.
Enter Workspace	Button	Yes	Transitions to main dashboard layout.

Screen 2: Dashboard / Factory Overview Screen

- **Purpose:** Provide a centralized view of factory metadata, metrics, and technical debt audit logs.
- **User Access:** All users.
- **Navigation:** Automatically loads at root (/) after setup completion.
- **Visual Description:** Divided into a statistics grid at the top, an insights alert panel, and an AI narrative box at the bottom.
- **Screenshot:**

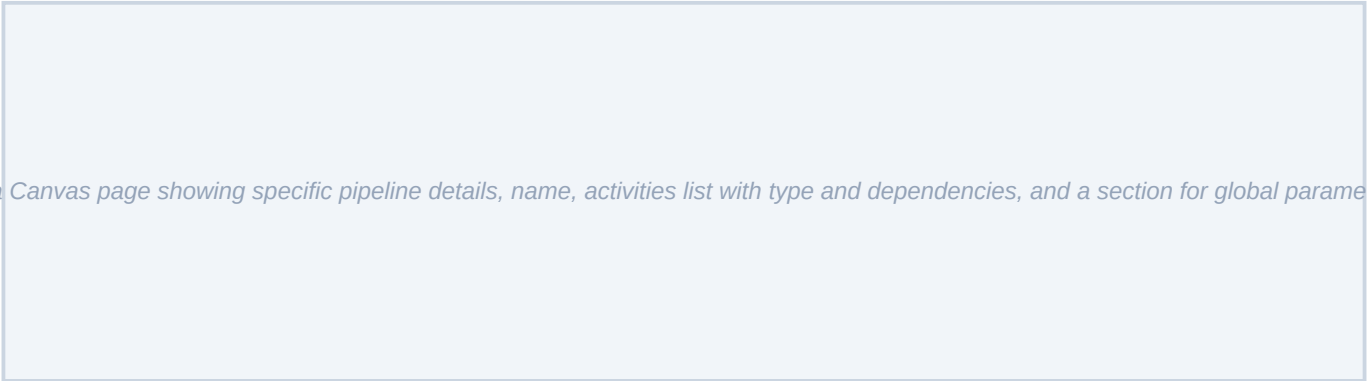


Showing statistics cards like Pipelines, Activities, Datasets, Linked Services, Triggers, Global Params, Data Flows, and Orphan Pipelines, plus AI Insights

- **Field Details:**
- **Stat Cards:** Pipelines, Activities, Datasets, Linked Services, Triggers, Global Params, Data Flows, Orphan Pipelines. Clicking a card opens a detailed drill-down table.
- **Factory Insights Alert Panel:** Displays colored warning alerts for Zero-retry activities, Orphan pipelines, Concurrency collisions, and Duplicate parameters.
- **AI Narrative Box:** Displays a natural language summary of the factory generated by the AI reasoning engine.

Screen 3: Data Canvas / Pipeline Detail Screen

- **Purpose:** Detail pipeline activity chains, parameters, variables, and links.
- **User Access:** All users.
- **Navigation:** Click on a pipeline row in the Dashboard table or sidebar list.
- **Screenshot:**

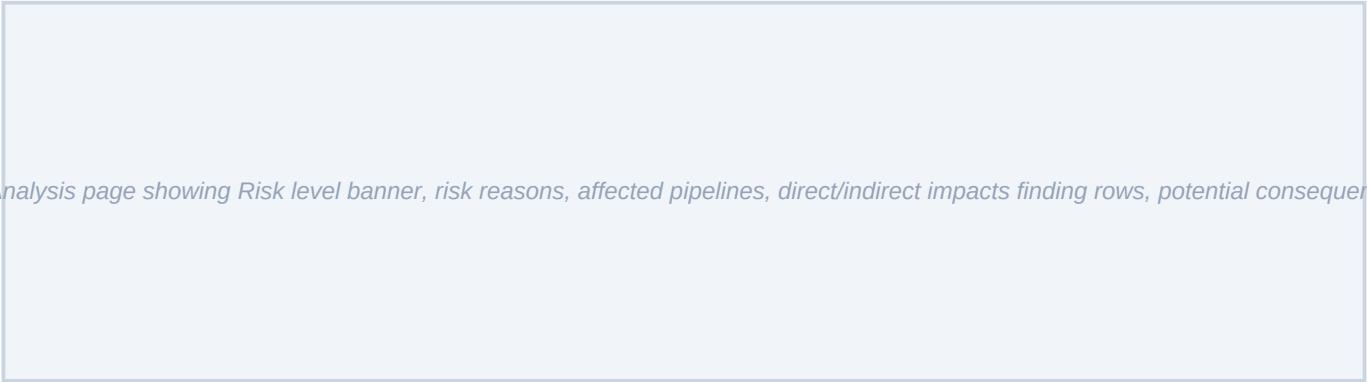


Folder: Data Canvas page showing specific pipeline details, name, activities list with type and dependencies, and a section for global parameters and variables

- **Field Details:**
- **Back Button:** Navigates back to the Dashboard table.
- **Impact Analysis Button:** Opens the Change Impact Panel for the pipeline.
- **Activities Table:** Lists name, activity type, and dependsOn array.
- **Parameters & Variables Box:** Summarizes pipeline-level parameters and variables.

Screen 4: Change Impact Panel / Blast Radius Screen

- **Purpose:** Run what-if simulations to calculate change risk.
- **User Access:** Data Engineer, Technical Lead.
- **Navigation:** Click "Impact Analysis" button from a pipeline detail screen.
- **Screenshot:**



Impact Analysis page showing Risk level banner, risk reasons, affected pipelines, direct/indirect impacts finding rows, potential consequences, and AI Recommendation Box.

- **Field Details:**
- **Change Type Select:** Dropdown to choose DELETE, REMOVE, DISABLE, MODIFY, REPLACE, RENAME, DECOMMISSION.
- **Analyze Impact Button:** Triggers the simulation.
- **Risk Banner:** Displays numerical score and level (CRITICAL, HIGH, MEDIUM, LOW).
- **Direct/Indirect Impacts list:** Displays expandable list rows of findings. Clicking a row reveals evidence, relationship type, and impact reasoning.
- **AI Recommendation Box:** Highlighted panel with remediation instructions.

Screen 5: AI Chat Sidebar & Settings Modal

- **Purpose:** Configure credentials and interact with the AI assistant.
- **User Access:** All users (settings restricted to Leads/Architects).
- **Navigation:** Gear icon on the left sidebar opens Settings; Chat sidebar is permanently visible on the left pane.
- **Field Details:**
- **Model Dropdown:** Selects Google Gemini, Azure OpenAI, NVIDIA NIM, or Mock.
- **Chat input:** Text area to type natural language questions.
- **Settings Modal Input:** Form fields for Google API Key, Azure OpenAI Endpoint, Azure OpenAI API Key, and NVIDIA API Key.

10. Business Rules and Validations

This section defines the key functional business rules and parameters enforced by PIE:

1. **Least-Privilege Reader RBAC:** PIE is strictly read-only. It never performs write actions on the Azure resource provider. The backend holds no API functions for editing pipelines.
 2. **PKCE State Validation:** Authentication requires matching PKCE `code_challenge` and `code_verifier`. Verification states are saved in-memory and deleted immediately upon authentication callback or after a 10-minute timeout.
 3. **Session TTL Validation:** User sessions are assigned a lifespan based on the token's `expires_in` property. Expired sessions are automatically cleared by the session store, forcing users to re-authenticate.
 4. **Context Compression Rules:** Raw JSON schemas must be stripped of coordinate data, workspace layout details, and GUIDs. The TokenBudgeter enforces a hard token limit (e.g. 1500 tokens) to ensure prompt sizes remain within API boundaries.
 5. **Intent Routing Rules:** Keyword mapping routes queries to the corresponding prompt template (e.g., "delete" -> IMPACT, "pyspark" -> CODE_GEN, "audit" -> SECURITY_AUDIT).
 6. **Deletion Simulation Scoring:** Risk scores are calculated dynamically:
 - **High Risk (80-100):** Breaking direct readers or writers that participate in business-critical copies.
 - **Medium Risk (40-79):** Modifying datasets with downstream pipelines that have retry policies.
 - **Low Risk (0-39):** Deleting unreferenced or orphan assets.
 7. **Schedule Concurrency Heatmap:** Schedule overlap is computed per hour. Hours with concurrent schedules exceeding the peak concurrency threshold (e.g. 5+ pipelines firing simultaneously) are highlighted in red.
 8. **Technical Debt Rules:**
 - A pipeline is an "orphan" if it has 0 incoming edges from schedule triggers or Execute Pipeline activities.
 - An activity is "fragile" if its retry count is explicitly set to 0.
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11. Notifications and Messages

PIE displays the following key user-facing system notifications and error messages:

- **Authentication Failure:** Displayed on the Setup Wizard if Microsoft Entra ID returns an authorization error.
 - *Triggering Action:* Failed login.
 - *Intended User:* All.
 - *Message:* "Auth failed: [Error reason]" (Red banner, user action: Retry login).
 - **Factory Syncing Message:** Displayed in the Setup Wizard during extraction.
 - *Triggering Action:* Selecting factory for sync.
 - *Intended User:* All.
 - *Message:* "Syncing [Factory_Name]... Extracting pipelines, mapping datasets, and building graph..." (Spinner widget).
 - **Workspace Ready Success:** Displayed when sync succeeds.
 - *Triggering Action:* Sync complete.
 - *Intended User:* All.
 - *Message:* "Ready to Explore! Your workspace is synced and ready." (Green checkmark, action: click Enter Workspace).
 - **API Key Missing Alert:** Shown in backend logs and triggers a warning badge in the Chat if a premium model is selected without a key.
 - *Triggering Action:* Running AI query with missing key.
 - *Intended User:* Lead / Architect.
 - *Message:* "[Provider] API key not found. Using Mock Provider instead." (Warning indicator, action: add key in Settings).
 - **Disambiguation Notice:** Displayed in the Change Impact Panel if multiple assets match the name.
 - *Triggering Action:* Deleting/modifying asset with non-unique name.
 - *Intended User:* Engineer / Lead.
 - *Message:* "Warning: Multiple assets match this name. [Detailed list of matches]" (Yellow warning banner).
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12. Reports, Downloads, and Outputs

PIE generates three primary outputs for users and stakeholders:

1. **Git-Ready Markdown Documentation:** Users can click a "Generate Specs" button inside the data canvas or chat to export a complete Markdown file detailing the pipeline. Contains description, activity chains, parameters, and dependency maps. Developers use this to commit documentation to Git repositories.
 2. **PySpark/dbt Migration Templates:** Generated in the AI Chat container when code migration requests are submitted. Provides drop-in code templates representing the data transformations of copy or mapping data flow activities, speeding up warehouse modernization.
 3. **Graph Topology JSON:** Developers can export the full in-memory knowledge graph ([GET /graph/topology](#)) as a JSON file representing the list of nodes and edges, allowing integration with custom lineage rendering tools.
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13. Error Handling and Edge Cases

Key system-level recovery strategies for user actions include:

- **Azure API Throttling or Outage:** If Azure ARM APIs rate-limit or fail, PIE degrades gracefully by serving the factory overview from the local disk cache (provenance "cache") and displaying a notification.
 - **AI Search or Foundry Offline:** If the LLM provider fails to respond within a timeout (e.g. 30 seconds), the AI assistant informs the user and falls back to rendering a deterministic lineage report generated by the graph traversal engine.
 - **Duplicate API Key Entries:** The settings router handles saving keys by reading the existing `.env` file, parsing keys into a dictionary, overwriting values, and writing the clean configuration back, preventing key accumulation.
 - **No Search Results:** Natural language searches that return no matches display a friendly message: "No matching assets found. Try searching by technology (e.g. CSV, Azure SQL) or folder."
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14. Assumptions, Limitations, and Open Points

- **In-Memory Cache Limitation:** The MetadataRepository is stored in-memory. Although real factories are saved to a local disk cache to survive restarts, session logins are lost when the server restarts.
 - **Authentication Callback Configuration:** Assumes that the local server can host and capture redirects on Port 8100. Local firewall configurations must permit traffic on port 8100.
 - **Key Vault Integration:** API keys entered in Settings are written to a local `.env` file in plaintext. In production, this must be integrated with Azure Key Vault.
 - **Multi-Factory Chat Scope:** The Chat Container scope is pinned to the currently active factory stored in the browser's `localStorage(selected_factory)`. Switching factories refreshes this context.
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15. Conclusion

The Platform Intelligence Engine (PIE) serves as a critical knowledge layer and engineering advisor for Azure Data Factory. By combining a deterministic knowledge graph with semantic AI reasoning, it provides data engineers, architects, and operations teams with real-time decision support.

The core user journey transitions smoothly from browser-based authentication to automated metadata ingestion, giving teams access to interactive data canvases, change blast-radius calculators, and automated governance reviews. PIE ensures that enterprise data platform changes are executed safely, rapidly, and with high confidence, preventing costly production outages and preserving engineering knowledge across the organization.